

Economic History II

Political identity and values

FGV EPGE

Outline of the class

- **Structural transformation and value change:** the British abolitionist movement. Figueroa and Fouka (2023).
- **Nation building through foreign intervention:** the US bombing of Vietnam. Dell and Querubín (2018).
- **The origins of gender roles:** women and the plough. Alesina, Giuliano and Nunn (2013).
- One question runs through all three: what makes a society's values change, and what keeps them fixed.

Structural transformation and value change

Figueroa and Fouka (2023)

The British abolitionist movement

Structural Transformation and Value Change: The British Abolitionist Movement

Valentín Figueroa and Vasiliki Fouka

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ABSTRACT

What drives change in a society's values? From Marx to modernization theory, scholars have identified a connection between structural transformation and social change. To understand how changes in a society's dominant mode of production affect its dominant values, we examine the case of the movement for the abolition of slavery in the late 18th and early 19th century Britain, one of history's most well-known campaigns for social change, which coincided temporally with the Industrial Revolution. We argue that structural transformation alters the distribution of power in society and enables groups with distinct values and weak economic interest in the status quo to mobilize for change. Using data on anti-slavery petitions, membership in abolitionist groups, MP voting behavior in Parliament and economic activity, we show that support for abolition was strongly connected to manufacturing at the aggregate and individual level. We rely on biographical data and the analysis of parliamentary speeches to show that industrialists were relatively less reliant on income from slavery and were characterized by a universalist worldview that distinguished them from established elites. Together, our findings suggest that both values and economic interest play a role in driving social change.

The puzzle

- Value systems, mappings from actions to moral valuations, vary across societies and over time.
- Footbinding was practiced in China for close to a millennium, then became stigmatized and faded in under a hundred years.
- Approval of same-sex relations among Americans went from 20% in 1990 to 60% in 2020.
- Britain, 1787 to 1833: slavery goes from a fundamental institution of the Empire to **morally repugnant**.

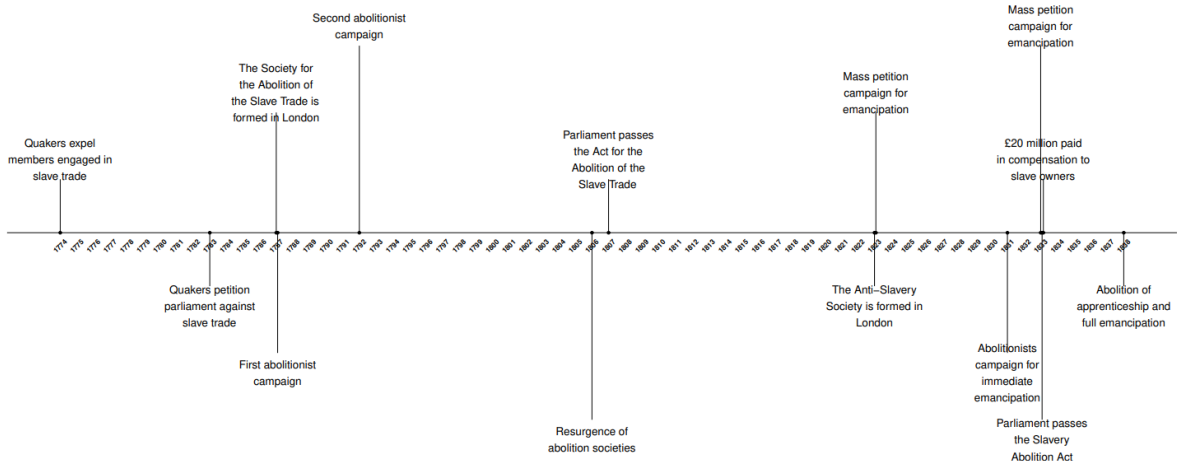
British slavery

- First decade of the nineteenth century: the Atlantic slave trade carried over **60,000** enslaved persons per annum.
- Britain was the dominant player. More than half of those transported between 1791 and 1805 went on British ships.
- Slave labor was the main input for sugar. British colonies in the West Indies, Jamaica and Trinidad among them, produced more than half the world's sugar in 1805.

The movement

- 1783: the **Quakers** put abolition on the parliamentary agenda, on religious grounds. Quickly dismissed.
- 1787-8: the Society for Effecting the Abolition of the African Slave Trade. Nine of the twelve founding members were Quakers.
- Method: printed material, public meetings, lecture tours, and above all bursts of petitions flowing to Parliament.
- Manchester, 1788: 11,000 signatures, a fifth of the town's population.
- 1778 to 1833: at least 12,124 petitions. In 1814, and again in 1833, 20 percent of the British adult population signed one.

Figure 1. Timeline of abolitionist movement



Reading the timeline

- Three stages of popular campaigning: against the slave trade, 1788 to 1807; for the living conditions of the enslaved, 1823; for emancipation, 1830 to 1833.
- Policy tracked mobilization. The slave trade ended in 1807, the enslaved in the West Indies became paid apprentices in 1833, emancipation came in 1838.
- Petitions filed before 1807 asked for abolition of the trade. All later ones asked for abolition of slavery itself.

Abolition was costly

- The 1833 bill compensated slave owners **20 million pounds**, about 5% of British GDP, only fully paid off in 2015.
- Kaufmann and Pape put the whole anti-slavery effort at roughly 1.78 percent of national income per year, 1808 to 1867.
- That figure includes the naval, legal and diplomatic cost of suppressing the trade, lost customs revenue, the lost sugar-carrying trade, and higher sugar prices at home.
- Also the loss of life, mainly from disease, among the Royal Navy crews enforcing the ban.

Industrialization, at the same time

- The Industrial Revolution begins in England around 1750: textiles, metallurgy, transportation.
- 1780 to 1870: real GDP per capita doubles, putting Britain 70 percent ahead of France and Germany.
- The share of the labor force in agriculture halves, to 22.7 percent. Around a third of the population works in manufacturing.
- Manchester, Birmingham and Sheffield grew more than **ten-fold** between 1750 and 1850.

Who the industrialists were

- Crouzet's sample of major industrialists, 1750 to 1850: 85% came from a middle-class background, almost half from low-middle-class families.
- In Crouzet's words, "shopkeepers, self-employed craftsmen and artisans, cultivators of various kinds".
- And: "neither the upper class nor the lower orders made a large contribution to the recruitment of industrialists".
- Often religious **dissenters**: a Protestant work ethic together with enlightened ideas about liberty. Josiah Wedgwood, Richard Reynolds.
- By the Victorian era England had two middle classes: a wealthier, capital-intensive one based on commerce in London, and a labor-intensive one based on manufacturing in the North.

Their standing

- Regional influence: coal and iron masters in South Wales, mill owners in Yorkshire and Lancashire, engineers and shipbuilders in Liverpool and Scotland. By the early Victorian period these men had founded dynasties.
- Manufacturers were also the arbiters of social relations with a growing industrial workforce.
- Nenadic: “In the eyes of the middle class, manufacturers possessed moral authority”.

The historiographical debate

- **Williams** (1944): capital from the slave trade financed early industry; then industry overshadowed sugar, the West Indies mattered less, and mercantilism became an obstacle to British exports.
- What is central in Williams is the relative, not the absolute, contribution of slavery to the British economy.
- **Davis** (1966): abolition instigated by the industrial bourgeoisie, slavery as the antithesis of the free-labor order on which their rise depended.
- **Drescher** (1986): a middle class phenomenon, artisans and miners excluded from the political system, petitioning as the extra-institutional route.

The argument

- Structural transformation, the rise of a new dominant sector, shifts the **distribution of power** in society.
- Change is initiated by ascending groups with three things at once: weak economic interest in the status quo, values distinct from entrenched elites, and enough power to act on them.
- The cost of mobilizing falls as participation rises. So once industrialists move, weaker groups with aligned values can follow.
- Both economic interest and values do work here. Neither alone gets the timing right.

The model

- Each group has values, income from slavery-related activity, income from everything else. Power is a function of income.
- A group mobilizes when its pro-reform values clear a threshold set by its slavery income plus the cost of taking part, divided by how much its own participation raises the chance of success.
- Agents are backward-looking: this period's odds depend on last period's participation.
- That gives growth that accelerates, and an order in which groups join.

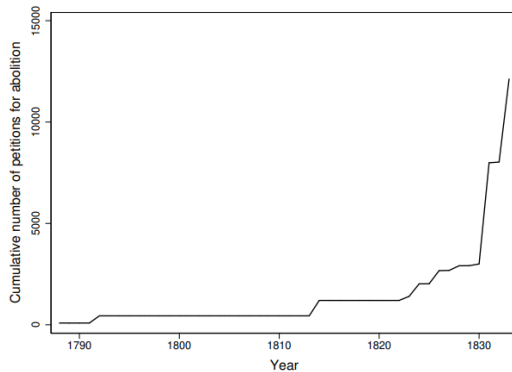
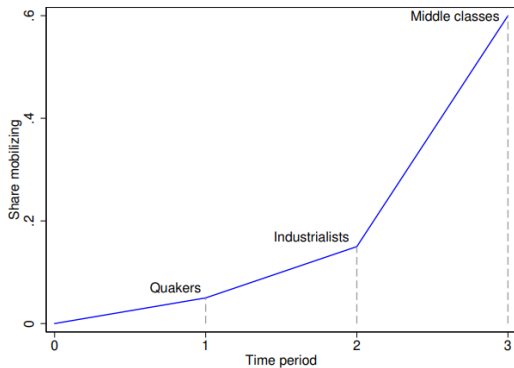
Table 1. Model parameters applied to abolitionism in Britain

Group	v_g	y_g^s	y_g^o	$p(y_g^o + y_g^s)$
Quakers	High	Low	Low	Low
Industrialists	Intermediate	Low	Low $\rightarrow$ High	Low $\rightarrow$ High
Middle classes	Intermediate	Low	Low	Low
Landed elite	Low	Intermediate	High	High
Planters	Low	High	Low	High

The five groups

- **Quakers**: extreme anti-slavery values, low slavery income, low power. They move first on values alone.
- **Industrialists**: intermediate values, low slavery income, power moving from low to high as industry grows.
- **Middle classes**: intermediate values, low slavery income, low power. They follow once the cost of joining has fallen.
- **Landed elite**: conservative values, some slavery income, high power. **Planters**: conservative values, high slavery income. Neither joins.

Figure 2. Dynamics of mobilization against slavery



Model and data, side by side

- Left: the share of the population mobilizing in each period under the model. Quakers, then industrialists, then the middle classes.
- Right: cumulative petitions against the slave trade and slavery submitted to the Commons, 1787 to 1833.
- The same shape. Few adherents in 1787, a national cause by 1833.

The data

- **Petitions**: topic, date and town of origin from the indices of the Journal of the House of Commons, geocoded to about 9,000 parishes in England and Wales.
- **Signatures**: the 1806 Manchester petition, one of the few documents to survive the 1834 fire at Westminster.
- **Prominent men**: everyone with a biography in the Oxford Dictionary of National Biography, matched to the 1787 Anti-Slavery Society's published subscriber list.
- **Press**: 12,421 articles mentioning slavery, 1787 to 1833, from the British Newspaper Archive.
- **MPs**: biographical dictionaries plus the Legacies of British Slavery database.



1787-1792



1806-1814



1821-1829



1830-1833

The spread of petitioning

- Each dot is a petition sent by a town to Parliament, in the four main stages of the movement.
- 1787-1792, then 1806-1814, then 1821-1829: scattered, and thickening.
- 1830-1833: England is covered.
- The abolitionist cause had few adherents in 1787 and was extremely popular at the national level by 1833.

Measuring industrialization

- No year-by-year local measure of industrialization exists for 1787 to 1833. The authors build two.
- **Coal**: the share of a parish over coal-bearing bedrock, multiplied by total coal consumption in England that year.
- **Water mills**: fifteenth-century mills from the Inquisitions post Mortem, multiplied by an index of national industrial output.
- Crafts and Wolf: “if there was a necessary condition for having a cotton textiles mill in 1838, a cheap source of power is the nearest candidate”.

Why two proxies

- Coal mattered as cheap energy for metalworking. Water mills capture water power and the mechanical skills of millwrights, which went into textiles.
- Water power still made up 70% of industrial power in 1800. Between 1760 and 1830 its importance doubled, while that of coal grew by a factor of 32.
- Both proxies predict the share of male workers employed in manufacturing in 1831.
- Each parish is compared with itself over time, so anything permanent about the place drops out.

Industrialization and anti-slavery petitions

- As a parish industrialized, it sent more petitions against the slave trade and slavery.
- Coal: parishes with the largest area share over coal deposits sent **19.4%** more petitions over the period than parishes with none.
- Water mills: 21% between parishes with none and parishes with the maximum of ten.
- Both hold when parishes are allowed onto different trends by terrain, soil and location, and once Quaker meeting houses and the local gentry are in.

The worry about medieval mills

- Parishes with fifteenth-century mills may have been different before the Industrial Revolution in ways that terrain, soil and location cannot capture.
- The direction is not obvious. Earlier development could mean faster urbanization and more petitioning.
- It could also mean less turnover in the social structure, closer ties to landed elites, and **more conservative** values.
- The fix: predict where a mill could have been built from the physical environment alone.

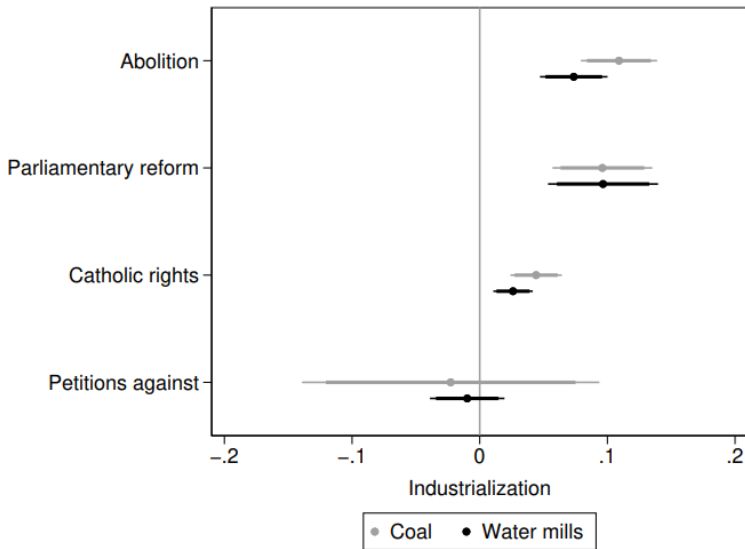
Water power potential

- Basic hydraulics: the power a wheel can generate depends on flow and on **head**, the elevation differential between where water enters and exits.
- Gordon (1983) approximates potential with the product of a river's flow and its fall at each point of a drainage basin.
- Here: average flow accumulation, the upstream area draining into a site, multiplied by terrain slope.
- Flow and slope each enter on their own. Only their product is used to predict where mills were.

What that exercise shows

- Parishes combining high flow with sloped terrain had water mills in the fifteenth century, and registered more anti-slavery petitioning as industrialization took off.
- The estimate obtained this way is an order of magnitude larger than the plain correlation.
- The reading the authors favor: parishes that had a mill only because their topography suited it are the ones most likely to have turned that mill into industry.
- The estimate barely moves when the full set of parish characteristics is added.

Figure 4. Industrialization and other petitioning campaigns



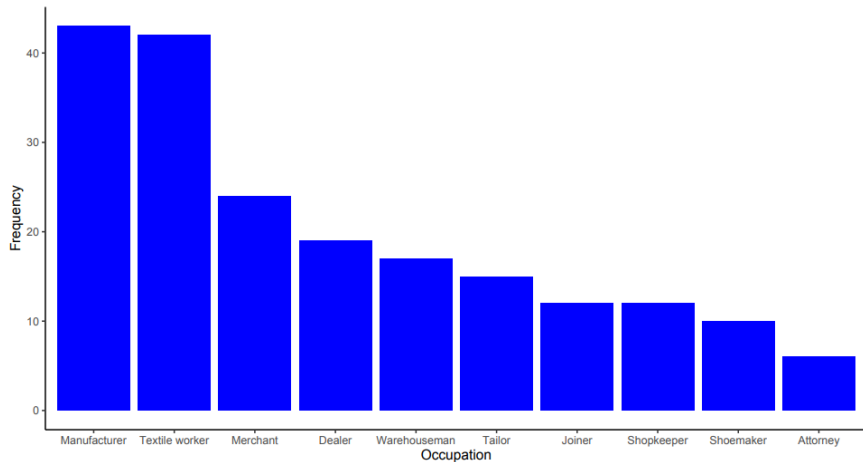
Not only abolition

- Same exercise, other petitioning campaigns of the period.
- Industrializing parishes petitioned more for **parliamentary reform**: a wider franchise, and seats moved from the rotten boroughs to towns that had grown with industry.
- And more for Catholic rights: repeal of the restrictions on Catholics and Dissenters, including on holding public office.
- Petitions *against* abolition show no positive relation to industrialization. So this is not mechanically population growth.

Who was actually petitioning?

- The parish-level pattern could be driven by the middling strata, or by the working class. Aggregates cannot tell them apart.
- The 1806 Manchester petition to the House of Lords: **2,348 signatures**, encouraged by Thomas Clarkson in reply to an anti-abolitionist petition.
- The Manchester and Lancashire Family History Society matched signatories to three trade directories: 312 unambiguous matches, 281 with an occupation listed.
- De Tocqueville on Manchester in 1835: “Here humanity attains its most complete development and its most brutish; here civilisation works its miracles, and civilised man is turned back almost into a savage”.

Figure 5. Top 10 most frequent occupations of Manchester petition signatories



The occupations

- **Manufacturers** are the most common occupation among the matched signatories: 43.
- Then textile workers, weavers, spinners and dyers: 42. Merchants 23, dealers 19, warehousemen 17, tailors 15.
- Manufacturers and textile workers signed in equal numbers, though textile workers were by far the larger group in the town.
- As a share of Manchester residents in each occupation in the 1813 census, industrialists signed at 7.8%, textile workers at 0.5%.

Signature order

- Signatures in Britain were not collected door to door. Petitions were posted in inns, shops and churches, and you walked there to sign.
- So the order in which someone signed says something about how central they were to the movement.
- The document is nine sheets of parchment stitched into a five-meter roll. Four proxies for order: page, column within page, entry within column, entry within page.
- Caveat the authors raise: notables were sometimes placed high on a list to induce others to sign. That is still consistent with the argument.

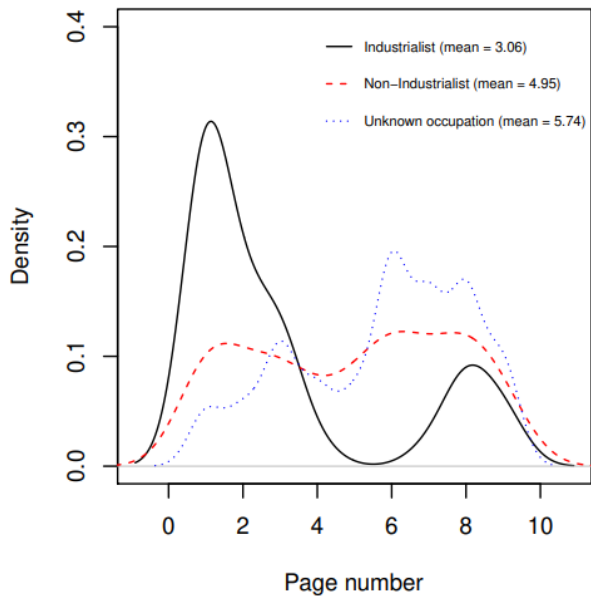
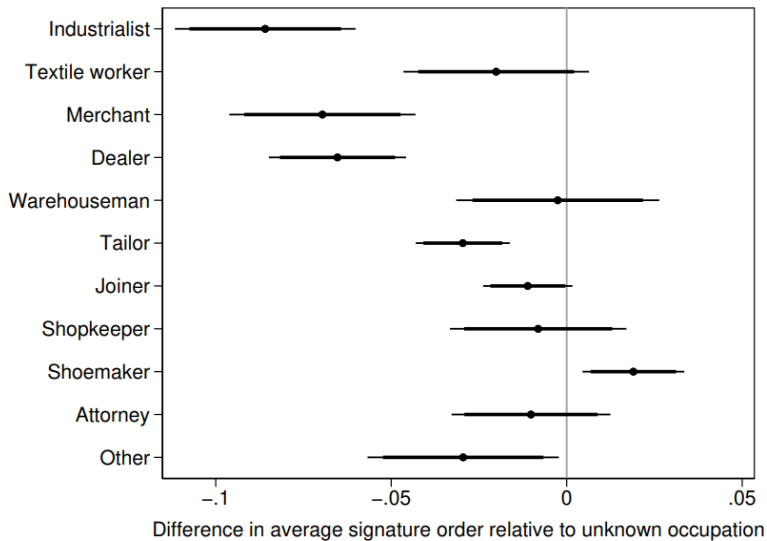


Figure 7. Effect of occupation on signature order



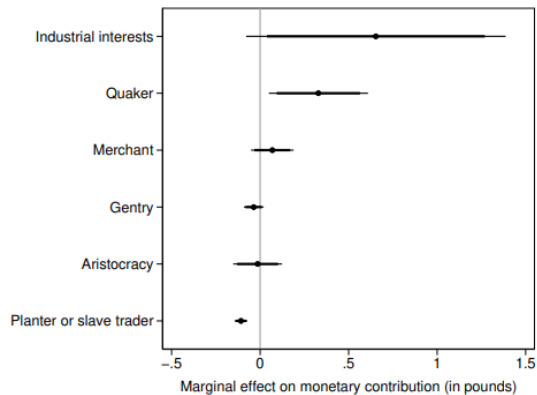
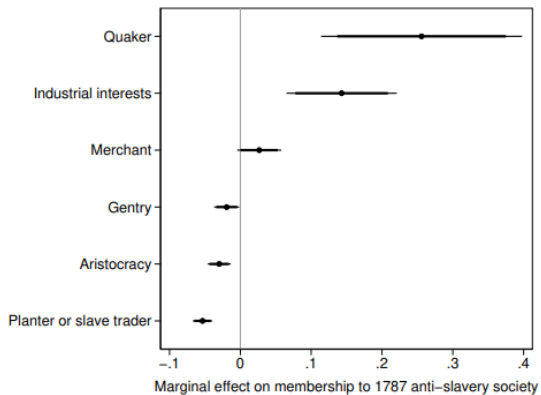
Manufacturers signed early

- Across all four measures the distribution for manufacturers sits to the left of the others.
- Page number: an average of 3.06 for industrialists, 4.95 for other known occupations, 5.74 for signatories whose occupation is unknown.
- By occupation, industrialists and merchants are the two categories most likely to sign first. Textile workers come later.
- Middle class elites head this campaign; the industrial workforce does not.

The Anti-Slavery Society of 1787

- 3,463 men have biographies in the Oxford Dictionary of National Biography and had reached adulthood by 1787.
- The Society published its full subscriber list, with each member's monetary contribution. **162** of the dictionary's men appear on it.
- Biographies coded for religion, social rank and economic interest: Quaker, aristocrat, gentry, industrialist, merchant, planter or slave trader.

Figure 8. Membership and contributions to the Anti-Slavery Society in 1787



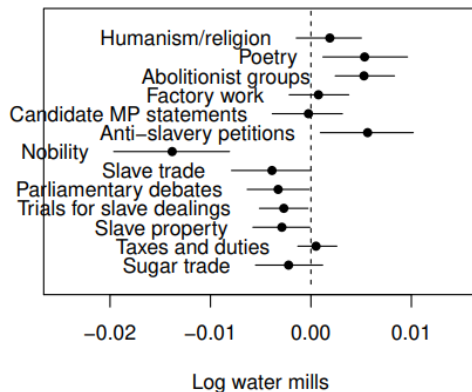
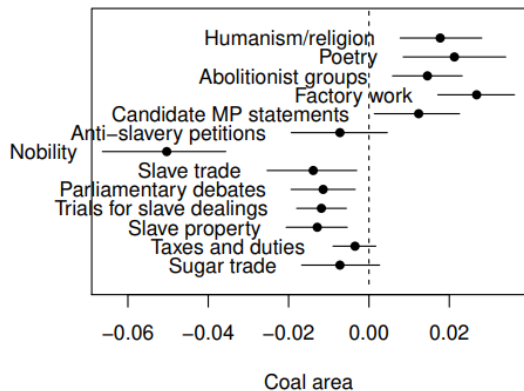
Who joined, and who paid

- Being a **Quaker** is the strongest predictor of membership among prominent men.
- Involvement in manufacturing comes next, and is the only other characteristic that predicts membership.
- On amounts contributed, manufacturers give the most, consistent with the highest income among groups holding anti-slavery values.
- Planters and slave traders join less and give less. The gentry and the aristocracy join less.

Values or interest? Read the press

- 12,421 articles mentioning slavery, published in 121 newspapers, placed at the level of the census registration district (625 districts).
- The authors sort the articles into 30 recurring themes and compare how prevalent each theme is in industrial and non-industrial districts.
- The question is whether slavery was discussed there in **moral** terms or in economic ones.

Figure 9. Topics of articles on slavery by location of publication



What industrial districts wrote about

- More prevalent in industrial districts: abolitionist groups, anti-slavery petitions, and the two topics the authors label Humanism/religion and Poetry.
- Less prevalent: slave property, the legal and financial side of owning people, and neutral references to the slave trade as commerce.
- Taxes and duties on sugar appear at similar rates everywhere.
- Not a matter of editorial line: newspapers in more industrial districts were not more liberal.

Humanism and religion

- April 1833, a letter from the Society of Friends reprinted in several papers: “Thou shalt love thy neighbor as thyself, under which term we believe are comprehended our fellow creatures of every nation, tongue and colour.”
- November 1832, a candidate MP’s statement: “The right of the slave to be raised from the level of the brute creation, to which he has been sunk through no fault of his own, and to be restored to that of human nature, in which he was created by his Maker”.

The topic we label “Poetry”, is dominated by poems against slavery, frequently imbued by the same humanist principles and condemning the institution as brutal and repugnant. An example from the Derby Mercury published in April 1823, is titled “Appeal to Britain – On the Slave Trade” and reads as follows:

Though thy name may be quoted again and again,
As a nation of freemen, a kingdom of men;

Yet a stain will be on thee, will tarnish thy name,
And a spot of dishonour will sully thy fame,
The blood of the slave thy fair honour shall stain.

This particular poem appealed to British exceptionalism as well as to a sense of honor, which scholars like Drescher (2002) and Appiah (2011) have identified as crucial for the change in attitudes on slavery.³²

Poetry

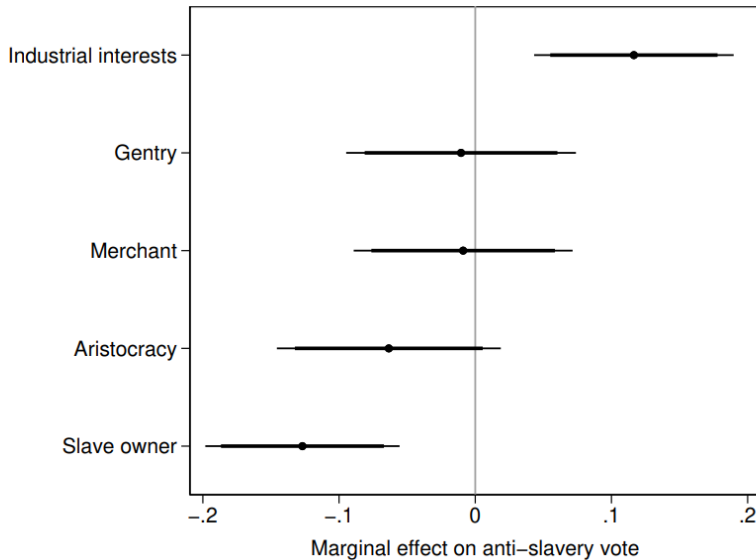
- The Derby Mercury, April 1823, under the title “Appeal to Britain”, on the slave trade.
- The appeal is to British exceptionalism and to a sense of **honor**, which Drescher and Appiah identify as crucial to the change in attitudes on slavery.
- Abolition was cast as a way for the Empire to display moral virtue after the loss of the American colonies.

Inside Parliament

The 1833 vote

- The 1832 reformed Parliament debated the Ministerial Plan for the Abolition of Slavery, enacted as the Slavery Abolition Act of 1833.
- An MP counts as anti-slavery if he voted at least once for limiting the apprenticeship period and for minimal compensation to planters.
- MPs with industrial interests were **36.2 percentage points** more likely to vote that way. About 27 points with the full set of individual and constituency characteristics.
- Slave owners voted against. Liberal and Radical leanings predicted support, as did Protestant dissent. Oxbridge, merchant standing, gentry and aristocracy did not.

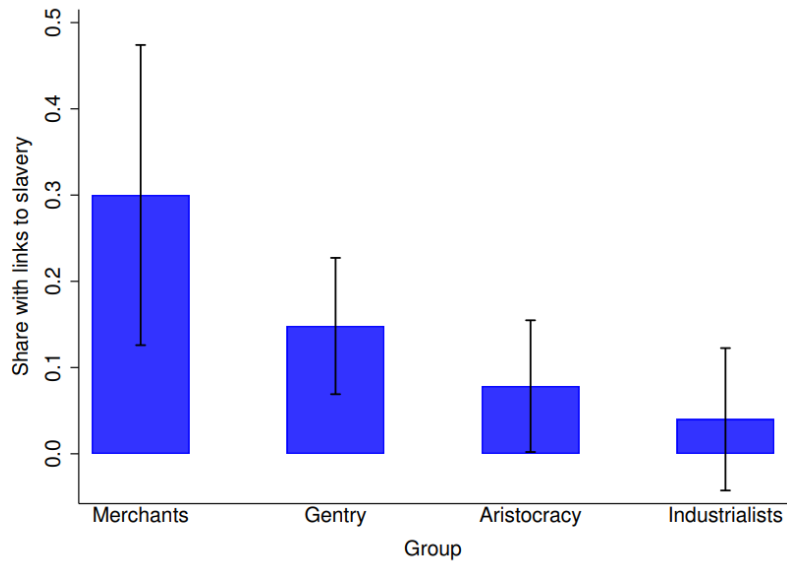
Figure 10. Support for abolition by group of MPs



The ranking of groups

- Industrialists the main supporters of abolition, slave owners its main opponents.
- The nobility less supportive, though the estimate is not quite sharp enough to call. Merchants and the gentry show no differential position.
- Earlier debates point the same way. In 1806-07, three of the four industrialist MPs who spoke supported abolition, against 30 of 46 other MPs.
- The one industrialist who opposed in 1806 was Sir Robert Peel, whose fortune came from cotton-spinning.

Figure 11. Links to slave income among British MPs



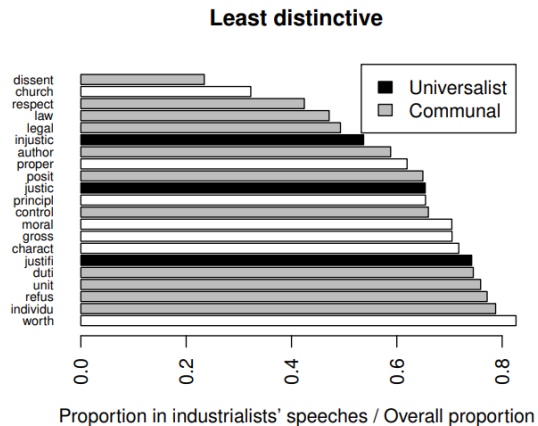
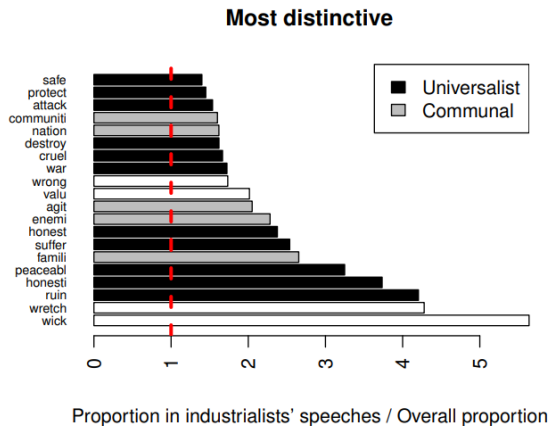
Direct links to slavery

- 51 of the 550 elected MPs appear in the Encyclopaedia of British Slave-ownership, as owners or otherwise linked to West Indian estates.
- Industrialists have the weakest links: **one** manufacturer in the entire list.
- Merchants 30%, the gentry 15%, MPs with a nobility title 7%.
- If slavery income were the whole story, merchants and gentry should oppose abolition more than the nobility. The votes show the opposite.

Measuring values from speeches

- Moral Foundations Theory partitions moral concerns into foundations, each with a different principle for what is right.
- **Universalist**: non-harm and care, fairness and reciprocity. **Communal**: ingroup loyalty, respect for authority and tradition.
- Universalism is not more altruism. It is altruism that declines less rapidly with social distance.
- Consistent with that: abolitionist manufacturers tolerated exploitation in their own factories, and abolitionists placed freed slaves near the bottom of society, a “grateful peasantry”.

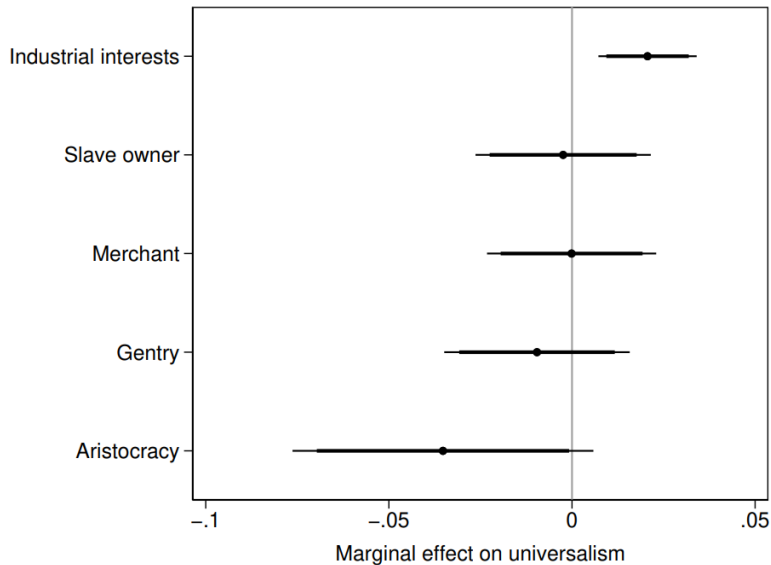
Figure 12. Words distinctive of industrial MPs



Words distinctive of industrialist MPs

- Distinctiveness: how much more often a word appears in industrialist speeches than in parliamentary speech on average.
- Most distinctive: safe, protect, cruel, wrong, suffer, honest, peaceable. The words characteristic of industrialists lean **universalist**.
- Least distinctive: church, respect, law, authority, duty, control. The words least frequent in their speeches, relative to overall use, reflect communal foundations.
- The corpus is 7,798 speeches from 1833-34, of which 368 were delivered by industrialist MPs.

Figure 13. Universalism by group of MPs



Universalism by group

- Industrialists clearly more universalist. The nobility clearly more communal. Gentry, merchants and slave owners in between.
- Not party, religion or education: neither Liberal MPs nor Protestant dissenters score higher.
- Not a topic effect either. It holds when slavery debates are dropped, and when each of 1,002 separate debates is accounted for on its own.
- Slave owners are no less universalist than the rest, which cuts against values simply adjusting to interest.

Universalism is not piety

- The dictionary has a separate category for words with moral content that belong to no foundation: righteous, moral, proper, correct. Industrial interests do not predict them.
- Purity words, tied to **sanctity and degradation**, are used far more by Protestant dissenters, and not by industrialists.
- That is the sanity check that a modern dictionary applied to 1833 speech is picking up something real.

Alternatives the authors rule out

- **Direct material gain.** Adam Smith called slavery inefficient: “A person who can acquire no property, can have no other interest but to eat as much, and to labor as little as possible”. But positions on free trade and the Corn Laws do not predict abolition votes.
- **Raw materials.** Textile manufacturers, dependent on slave-grown cotton, vote no differently from other industrialists.
- **Constituency pressure.** Neither petitioning in the constituency nor running unopposed changes anything. At least 130 elected MPs pledged to abolish in the anti-slavery journal *The Tourist*, and industrial interests do not predict who pledged.
- **Deflecting grievance.** Labor unrest and the Swing riots relate to anti-slavery votes in general, but no more so for industrialists.

Where the industrialist MPs came from

- 25 MPs with industrial interests against 521 others.
- Much less likely to be Oxbridge graduates, 8% against 37%. More likely to be Protestant dissenters, radicals, and supporters of free trade.
- No industrialist MP was a slave owner, and none held an aristocratic title.
- Consistent with Crouzet: manufacturers came from the middle classes and had weak links to the established aristocracy.

How elite values might have spread

- Mokyr: “Cultural entrepreneurs change what people believe, and if enough important people are converted, they will change institutions to conform with the new beliefs”.
- Abolition was the institutional change that condemned the old state of affairs as immoral and unjust.
- Prestige-based transmission: high-status individuals are imitated more often. Davis called the result ideological hegemony.
- Quirk and Richardson add a workplace channel: employers promoting sobriety, time discipline and personal responsibility alongside anti-slavery.

Conclusions

- Support for abolition was strongly connected to manufacturing, at the aggregate and at the individual level.
- Opposition came from two directions: high reliance on slavery income among planters, conservative values among the old aristocracy.
- The framework accommodates Williams, Davis and Drescher rather than choosing among them.
- The open question the authors leave: **why** industrialists were more universalist than everyone else.

The same question elsewhere

- Masera and Rosenberg: the loss of comparative advantage in cotton drove the decline of pro-slavery attitudes in the US antebellum South.
- Seyler and Silve: electoral support for emancipation in **Brazil** depended on how far a district relied on slavery against immigration as an alternative source of labor.
- Brazil abolished in 1888, last in the hemisphere, and paid no compensation to owners. Britain paid 5% of GDP.

Nation building through foreign intervention

Dell and Querubín (2018)

The US bombing of Vietnam

NATION BUILDING THROUGH FOREIGN INTERVENTION: EVIDENCE FROM DISCONTINUITIES IN MILITARY STRATEGIES*

MELISSA DELL AND PABLO QUERUBIN

This study uses discontinuities in U.S. strategies employed during the Vietnam War to estimate their causal impacts. It identifies the effects of bombing by exploiting rounding thresholds in an algorithm used to target air strikes. Bombing increased the military and political activities of the communist insurgency, weakened local governance, and reduced noncommunist civic engagement. The study also exploits a spatial discontinuity across neighboring military regions that pursued different counterinsurgency strategies. A strategy emphasizing overwhelming firepower plausibly increased insurgent attacks and worsened attitudes toward the U.S. and South Vietnamese government, relative to a more hearts-and-minds-oriented approach. *JEL Codes:* F35, F51, F52

Never before did the people of Vietnam, from top to bottom, unite as they did during the years that the U.S. was bombing us. Never before had Chairman Ho Chi Minh's appeal—that there is nothing more precious than freedom and independence—gone straight to the hearts and minds of the Vietnamese people.

—Tran Quang Co

The solution in Vietnam is more bombs, more shells, more napalm.

—General William DePuy

The war

- Geneva Accords, 1954: Vietnam divided at the 17th Parallel until nationwide elections in 1956. The elections were never held, in part because the US concluded Ho Chi Minh would win by a landslide.
- A Soviet-backed communist state in the north, the US-backed Ngo Dinh Diem in the south, and a communist insurgency in the south led by the **Viet Cong**.
- 1965: Johnson deploys around 200,000 troops. Peak of over half a million in 1969. Withdrawal in January 1973.
- The Vietnamese government estimates more than 3 million deaths between 1954 and 1975, over 2 million of them civilians. A 2008 study puts the toll at 3.8 million.

Two strategies

- **Overwhelming firepower**: air strikes, artillery, search-and-destroy raids. Reduce insurgent forces, disrupt operations, crush morale.
- **Hearts and minds**: security by embedding soldiers in communities, development aid, small units, limited firepower.
- Both were considered. In practice firepower dominated. Westmoreland described his strategy in one word: firepower.
- General DePuy: “The solution in Vietnam is more bombs, more shells, more napalm”.

Why firepower was supposed to work

- Deterrence. McGeorge Bundy told Johnson a reprisal policy “will set a higher price for the future upon all adventures of guerrilla warfare, and it should therefore increase our ability to deter such adventures”.
- Compliance. Walt Rostow argued that countering communism required “a ruthless projection to the peasantry that the central government intends to be the wave of the future”.
- When it did not end quickly, Westmoreland’s explanation: “the Oriental doesn’t put the same high price on life as does the Westerner. Life is plentiful, life is cheap in the Orient”.

The skeptics

- Insurgents are hard to locate. If the ordnance lands on civilians instead, it creates grievances that recruit for the other side.
- It can also widen the **legitimacy gap** between the insurgents and the South Vietnamese government.
- James Scott: a coercion-oriented approach is ill-suited to gaining cooperation, because citizens have many ways to undermine a state they do not genuinely support.
- Tran Quang Co: “Never before did the people of Vietnam, from top to bottom, unite as they did during the years that the U.S. was bombing us”.

The scale

- More than twice as many tons of explosives as in the Second World War. Four times more on South Vietnam than on the North.
- About **500 pounds of ordnance per inhabitant** of South Vietnam, equal in power to 640 Hiroshima-sized atomic bombs.
- 3.4 million combat sorties in Southeast Asia between 1965 and 1972, around 125,000 of them by B-52s. Around 400,000 tons of napalm.
- The guava cluster bomb carried 650 bomblets of 300 steel pellets each, 200,000 fragments per bomb.
- In the study's sample, 21% of hamlet-months had ordnance dropped within 5 kilometers of the hamlet centroid.

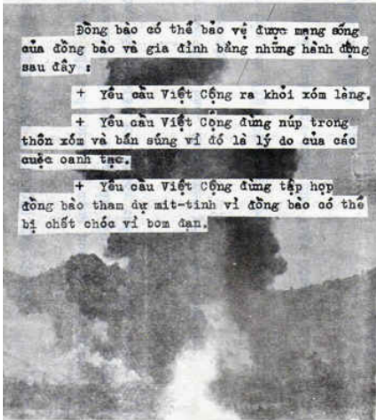
The rules of engagement

- Directly targeting civilians violated the laws of war. The rules still allowed many situations in which civilians could be hit.
- If any fire came from the vicinity of a village, it could be attacked without warning.
Free-fire zones: anyone could be fired on.
- Warnings came as blanket leaflet drops over a district, too general for villagers to tell whether they were on a target list.
- The common F-105 bomber had a circular error probability of 447 feet: half the bombs fell within that radius of the target. Enough to hit a village, not enough to hit an insurgent rather than a civilian.



Đồng bào có thể bảo vệ được mạng sống của đồng bào và gia đình bằng những hành động sau đây :

- + Yêu cầu Việt Cộng ra khỏi xóm làng.
- + Yêu cầu Việt Cộng đừng núp trong thôn xóm và bắn súng vì đó là lý do của các cuộc oanh tạc.
- + Yêu cầu Việt Cộng đừng tập hợp đồng bào tham dự mít-tinh vì đồng bào có thể bị chết chóc vì bom đạn.



Leaflet No. HQ-18-67 text:

To The People Who are in the Areas Temporarily Occupied by the Viet Cong

It is regrettable that the Government of Vietnam has to use bombs and artillery to drive the Viet Cong from places where they're hiding. In order to liberate your area, sometimes there is no other means. To protect your lives the government asks you to follow these measures:

- Do not live close to where the Viet Cong are gathering.
- Do not attend meetings hosted by the Viet Cong.
- Do not work for the Viet Cong.

Dear citizens. You can protect you and your family by taking the following actions:

- Ask the Viet Cong to leave the village.
- Ask the Viet Cong to stop hiding in the village and firing weapons; that is the reason for Government bombing.
- Ask the Viet Cong not to gather people for meetings because they may get killed by bombs and artillery.

The leaflets

- One leaflet shows a village leveled by bombing and tells citizens: “if you support the Viet Cong, your village will look like this”.
- “The Viet Cong hide among the innocent women and children in your villages to fire upon troops and aircraft. If the Viet Cong in this area use you or your village for this purpose, you can expect death from the sky.”
- Another is titled “Your village has been bombed” and explains that the village helped the VC in some way.
- The instruction to unarmed civilians was to make armed insurgents leave.

son were never in danger. As Viet Thanh Nguyen wrote in his Pulitzer Prize–winning novel: “We are all guilty until proven innocent, as even the Americans have shown. Why else do they believe everyone is really Viet Cong? Why else do they shoot first and ask questions later?” (Nguyen 2015, loc. 2902). Turse (2013, p. 55) argues:

At every turn, the onus was put on Vietnamese civilians to actively demonstrate that they were indeed noncombatants... by staying out of off-limits areas (the borders of which they might not know)... by not running or not walking in a certain way, or not standing still and thus looking unnatural; by somehow forcing armed guerrillas from their villages but also not carrying weapons, which would automatically brand a Vietnamese as VC.

How this looked from below

- Turse's summary is on the previous slide: the onus fell on Vietnamese civilians to demonstrate that they were noncombatants.
- Viet Thanh Nguyen: "We are all guilty until proven innocent, as even the Americans have shown. Why else do they believe everyone is really Viet Cong?"
- In a survey of generals who commanded in Vietnam, only 19% said the rules were "carefully adhered to throughout the chain of command" before My Lai became public. 15% said the rules were not "particularly considered in the day to day conduct of the war".

The question, and why it is hard

- Did bombing near population centers weaken the insurgency or strengthen it? And did it build a state that could stand on its own?
- Military forces target places where insurgency is already on the rise, so a raw comparison of bombed and unbombed places is uninformative.
- An unconditional lottery would be unfeasible, unethical, and also uninformative: resources go where they are believed to work.
- What is needed is variation **at the margin**, among places that had all been deemed potential targets.

McNamara and the whiz kids

- McNamara brought operations research from Ford to the Department of Defense in 1961 and surrounded himself with Rand analysts.
- Field data were key-punched into mainframes in Saigon and Washington and used to allocate resources.
- The tapes would have been destroyed. They survive because they were **subpoenaed** during a lawsuit against IBM.
- Much of the study's outcome data comes off those tapes.

The Hamlet Evaluation System

- 1967: US district advisers graded hamlets A to E on their own perceptions. Two 1968 studies found the grades did not track actual conditions.
- The Revised system, from July 1969: **169** monthly and quarterly questions on security, politics and economics, collected by advisers together with South Vietnamese district and province heads.
- Answers grouped into 19 submodels. Bayes's rule turns the answers in each submodel into a continuous score from 1 to 5.
- Nothing was collected for hamlets the VC controlled, about 3% of hamlet-months.

Figure A-3: Conditional Probability Matrix

ENEMY POLITICAL ACTIVITY	HMB05-0	.882	.788	.528	.571	.770
	HMB05-1	.068	.140	.314	.286	.152
	HMB05-2	.034	.070	.157	.143	.076
	HMB06-0	.891	.797	.510	.557	.764
	HMB06-1	.070	.134	.324	.294	.156
	HMB06-2	.035	.067	.162	.147	.078
	HMB07-0	.890	.785	.564	.500	.781
	HMB07-1	.055	.105	.215	.250	.110
	HMB07-2	.044	.084	.172	.200	.088
	HMB07-3	.011	.021	.043	.050	.022
	HMB08-0	.924	.858	.617	.278	.185
	HMB08-1	.048	.100	.260	.342	.350
	HMB08-2	.024	.038	.120	.378	.462
	HQB01-0	.502	.273	.079	.011	.001
	HQB01-1	.380	.469	.284	.074	.028
	HQB01-2	.104	.213	.476	.315	.096
	HQB01-3	.011	.026	.142	.440	.265
	HQB01-4	.001	.017	.018	.160	.609
	VQB01-0	.406	.244	.066	.010	.001
	VQB01-1	.450	.533	.277	.097	.018
	VQB01-2	.122	.183	.518	.272	.085
	VQB01-3	.018	.032	.120	.416	.244
	VQB01-4	.001	.003	.015	.203	.650

Where the probabilities came from

- Each question carries a conditional probability matrix: how likely each answer is if the hamlet is an A, a B, and so on.
- Field officers were surveyed: “you have been selected to participate in the design of a Bayesian processor”.
- The returned probabilities had high variance. Answers more than two standard deviations from the mean were dropped and the rest averaged.
- The same question entering two submodels can carry different probabilities. The details are arbitrary; the resulting score still tracks the security situation.

The rounding

- Each submodel's continuous score was **rounded to the nearest whole number** before anything else happened to it.
- 4.4999 becomes a 4/B, somewhat secure. 4.5001 becomes a 5/A, very secure.
- Combinatorial logic then aggregates the rounded submodel scores, two or three at a time, into one hamlet security score.
- Only that final score was printed and sent to military planners. The continuous scores were never printed or saved from the mainframe's memory.

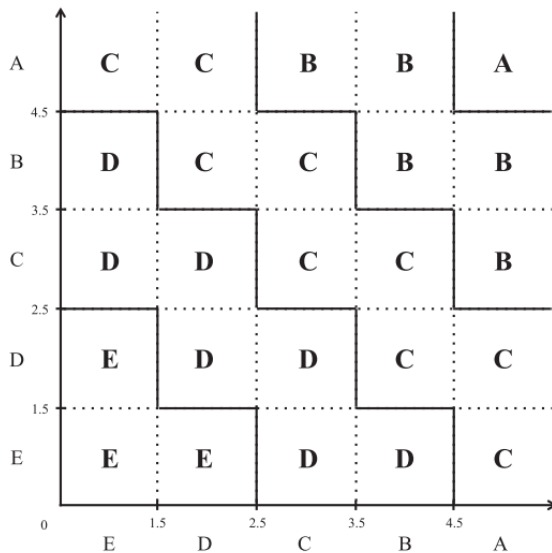
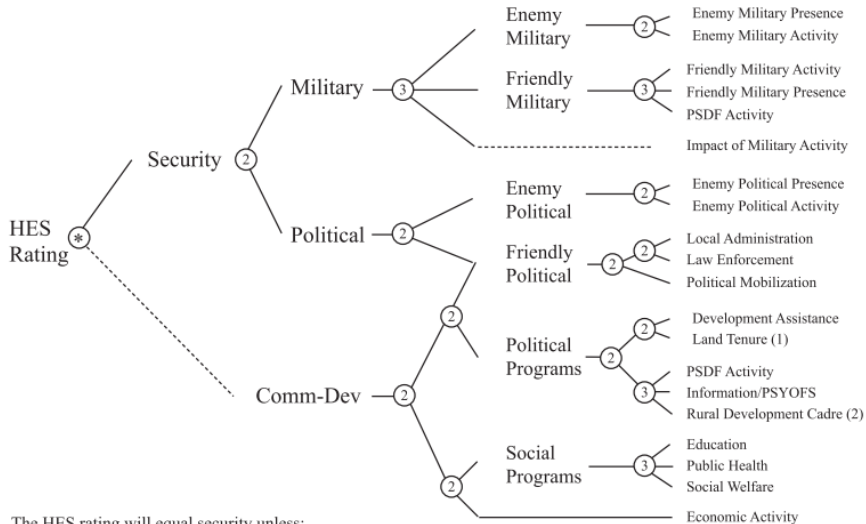


FIGURE I
Decision Logic (Two-Way)

This figure shows the aggregation logic for combining two submodel scores at a time.

Reading the decision logic

- The two axes are two submodel scores. Each cell is the resulting output grade. Thick lines are the thresholds between output grades.
- The two-way logic is symmetric: take the average of the two rounded grades and round down.
- A hamlet with 4.7 and 4.49 gets a B. A hamlet with 4.7 and 4.5 gets an A.
- Two hamlets with almost identical underlying conditions can land on opposite sides of a thick line.



The HES rating will equal security unless:

1. HMBO1 = 1 or 2 when the HES Rating will be set to 'V'

2. Community Development (Comm-Dev) is three letter grades higher or lower than security whereupon it will be raised or lowered by one letter grade as applicable. If security is Not-evaluated (N) , the HES Rating will = 'N'

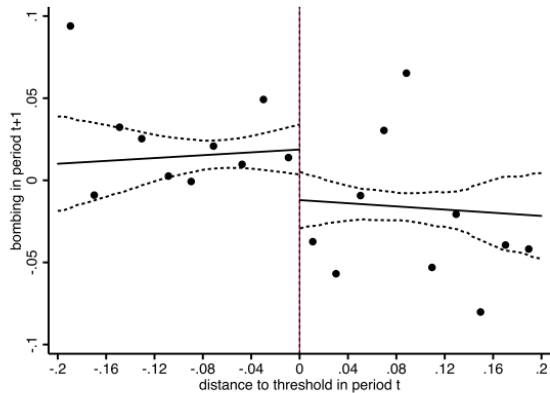
(1) If Land Tenure = 'N', use Development Assistance only.

(2) If RD Cadre = 'N', use a two way decision table rather than the three way table.

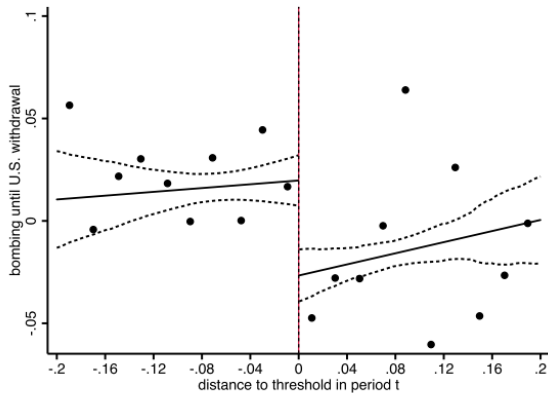
From 169 questions to one grade

- The 19 submodels are combined two or three at a time, up a tree, into a single score.
- The authors recompute the continuous scores: the conditional probability matrices were found in uncatalogued documents at Fort McNair, the question responses on tapes at the National Archives.
- The running variable is the distance from each hamlet to the nearest threshold, in the 19-dimensional space of continuous scores.
- Being **below** a threshold is what predicts being bombed.

(A) Immediate First Stage



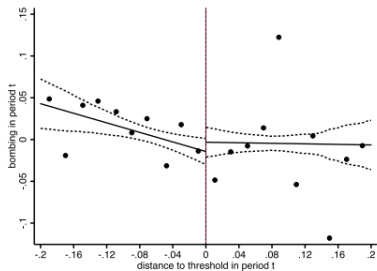
(B) Cumulative First Stage



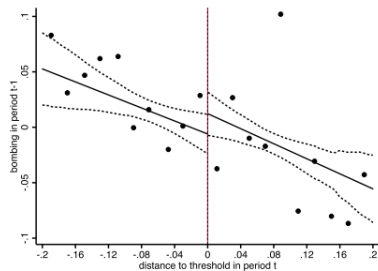
Rounded down, then bombed

- Horizontal axis: distance to the nearest rounding threshold. Negative means the hamlet was rounded down.
- Vertical axis: strikes near inhabited areas, in the following quarter on the left, cumulated to US withdrawal on the right.
- Strikes drop at the threshold. Being rounded down made a hamlet more likely to be hit.
- Around the A-B threshold, low As had about a 5% chance of being hit in a given month, high Bs about 10%. Around D-E, low Ds about 55%, high Es nearly 75%.

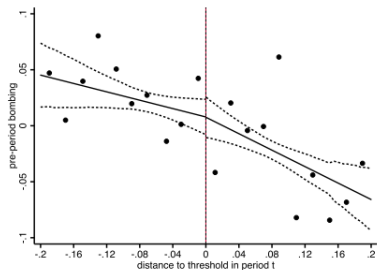
(A) Contemporaneous Bombing



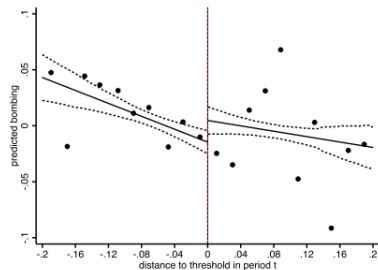
(B) $t - 1$ Bombing



(C) All Prior Quarters Bombing



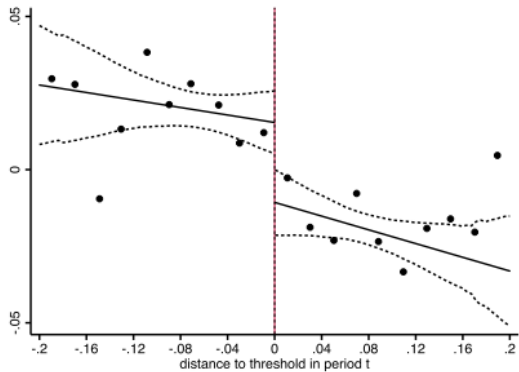
(D) Predicted Bombing ($t + 1$)



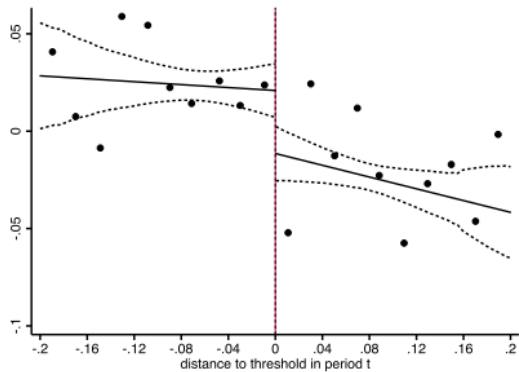
The placebos

- The data feeding the score arrived only at the close of the quarter, so there should be no contemporaneous effect. Panel A: smooth.
- Strikes in the previous quarter, and averaged over the whole period before: also smooth.
- Strikes predicted from the variables that enter the score but not from the score itself: smooth.
- In 1969 the score was computed but not disseminated. In that pilot year, being below the threshold did nothing.

(A) *VC Presence (Cumulative)*



(B) *Active VC Infrastructure (Cumulative)*



Bombing raised insurgent activity

- Moving from no strikes to the sample average raised the probability of an armed VC presence by 15 percentage points, from a mean of 0.19.
- The probability of an active **village guerrilla squad** rose by 27 points, from a mean of 0.38. That squad consists of locals, so this is not a reallocation of insurgent troops.
- The VC political branch, the VC infrastructure, was active 25 points more often.
- VC-initiated attacks on local security forces, officials and civilians rose by 9 points, from a mean of 16 percent of hamlet-months.
- VC extortion of residents rose by 23 points, from a mean of 0.27.

Why it could go this way

- Rank-and-file VC were part-time fighters operating near their own villages, indistinguishable from civilians without close local cooperation.
- Grievance. One soldier, after an American bomb killed his fiancée in Haiphong: he decided he had to go south and fight.
- Economic destruction lowers the opportunity cost of joining the insurgency.
- RAND interviewed 2,400 VC defectors and prisoners. Volunteers were more likely than forced draftees to report grievances against the government and economic hardship.

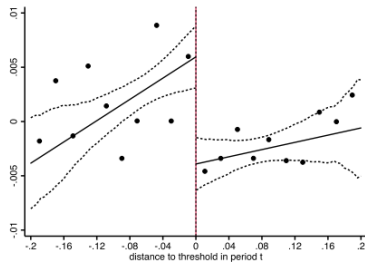
The loop

- The initial deterioration in security entered the next quarter's score, lowering it and raising the chance of being bombed again.
- That is why the cumulative effect is strong: bombing worsens security, which lowers the score, which brings more bombing.
- Ground troops followed. Friendly forces operated nearby 17 points more often over the course of the war, and US-initiated attacks rose.
- Spillovers to neighboring hamlets are limited, and where they exist they run in the same direction as the direct effects.

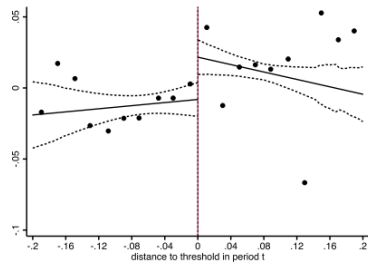
The nation-building case

- The point of the war was a state that could contain communism after the US left.
- **Forced-draft urbanization.** Huntington: “It is instead forced-draft urbanization and modernization which rapidly brings the country in question out of the phase in which a rural revolutionary movement can hope to generate sufficient strength to come to power.”
- Population growth in bombed hamlets is not measurably different. Vietnamese often stayed even when their towns were destroyed.
- One peasant, on those who remained: “Those who stayed didn’t have a choice”.

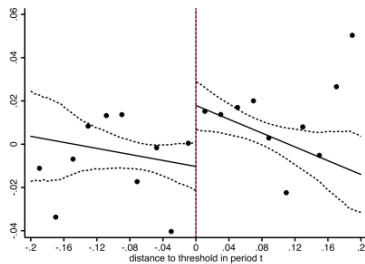
(C) *U.S. Initiated Attacks (Cumulative)*



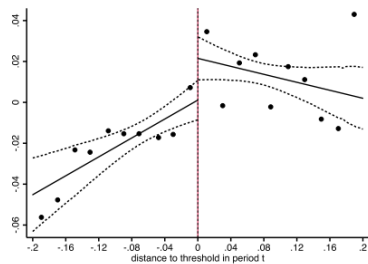
(D) *Village Committee Filled (Cumulative)*



(E) *Access to a Primary School (Cumulative)*



(F) *Participation in Civic Orgs (Cumulative)*



What bombing did to the state

- The probability that the local government systematically collected **taxes** fell 25 points, from a mean of 0.70.
- The village committee, which administered public goods, had all its positions filled 21 points less often, from a mean of 0.84.
- Access to primary school fell 16 points. Secondary school fell too.
- Household participation in civic organizations fell 13 points, from a mean of 0.29.
- Scott's point, in the data: citizens have many ways to undermine a state they do not support, well short of joining an insurgency.

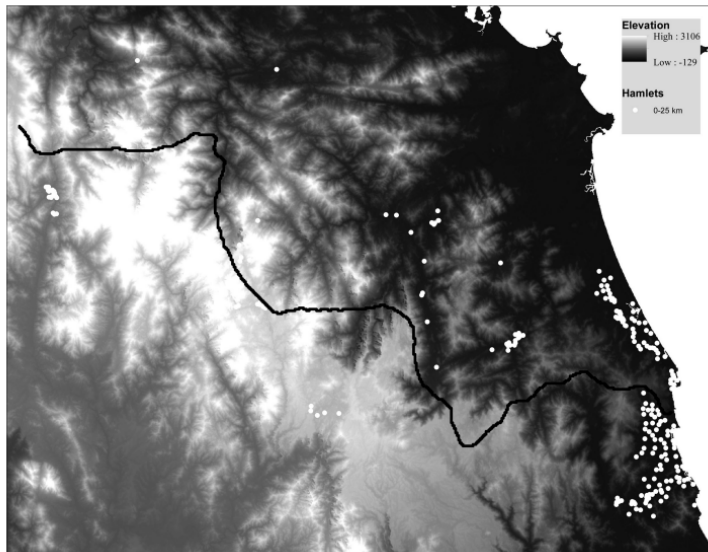


FIGURE VI
Corps Region Boundary

Army and Marines

- Four military regions. Corps I was commanded by the **Marines**, neighboring Corps II by the **Army**. The boundary followed a pre-existing province boundary.
- Army doctrine, from an official publication: “The only way to overcome VC control is by brute force”.
- Marine doctrine, from the 1962 manual: “a positive program of civil assistance must be conducted to eliminate the original cause of the resistance movement”.
- In practice: “Marine units built schools, roads, marketplaces, and hospitals, provided regular medical care, and provided training and equipment to local and regional militias”.

Are the two sides comparable?

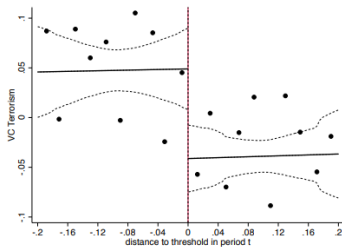
- VC attacks in the period before the Marines arrived, urbanization, elevation and slope are all balanced at the boundary.
- Colonial landmarks from geo-referenced 1929 maps, factories, markets, telegraphs and stations, are balanced. Military posts and total roads are higher on the Marine side.
- Armed Forces Qualifying Test scores of Army and Marine soldiers do not differ, nor does the racial composition of casualties.
- Boot camp lengths converged in 1967. Rotation was 12 months for the Army, 13 for the Marines.

What changes at the boundary

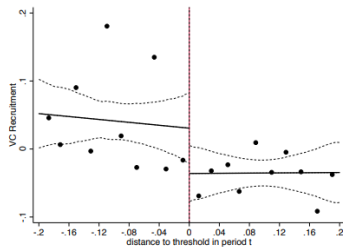
- The public goods the Marines targeted: high education provision 24 points more likely, high health care provision 56 points more likely.
- Less armed VC presence, fewer VC-initiated attacks on hamlets, fewer VC attacks on troops. No difference in the VC political infrastructure.
- Bombing was 11 points lower on the Marine side, though too noisy to call.
- Long's separate natural experiment, when the Army took over districts from the Marines: 88 percent of Army rounds were fired without a specific target, and refugees were produced five times faster.

Figure A-44: Public Opinion Data

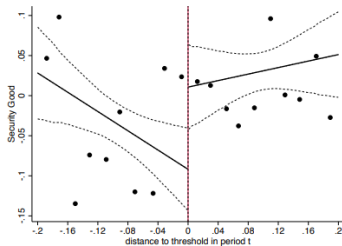
(a) *VC Terrorism (Cumulative)*



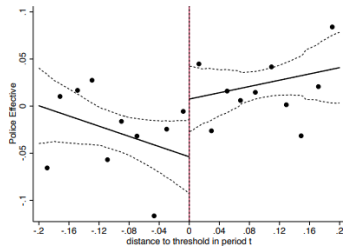
(b) *VC Recruitment (Cumulative)*



(c) *Local Officials Ensure Security (Cum)*



(d) *Police Effective Against VC (Cumulative)*



Attitudes

- Public opinion surveys, comparing respondents within 100 kilometers of the corps boundary.
- On the Marine side: 16 points more likely to say they liked Americans, 39 points more likely to report no hostility toward the United States in their community, 38 points more likely to call the US presence beneficial.
- More confidence in the South Vietnamese government, and better ratings of the ARVN, the regional forces, the police, and local officials.
- Consistent with hearts and minds being won, or at least lost less.

Conclusions

- Bombing increased VC military and political activity, weakened local government administration, and lowered noncommunist civic engagement.
- The mechanisms the authors point to are grievances and a widening of the political legitimacy gap.
- Kennedy, in 1954: “I am frankly of the belief that no amount of American military assistance in Indochina can conquer an enemy which is everywhere and at the same time nowhere”.
- McNamara, in private: “this [expletive] bombing campaign, it’s been worth nothing, it’s done nothing”.

A note on the advisers

- Bundy, a political scientist and once the youngest dean in Harvard's history, and Rostow, an MIT economist called America's Rasputin, were Johnson's national security advisers.
- The authors' reading: they used models of universal economic development and game-theoretic conflict to argue what *ought* to happen, and gave little thought to what might happen from forces outside those models.
- McNamara changed his mind as unfavorable evidence accumulated from 1966, and was marginalized for having no creative ideas about how to win.
- The paper closes on the price of policy advice that is not subjected to empirical critique.

The origins of gender roles

Alesina, Giuliano and Nunn (2013)

Women and the plough

ON THE ORIGINS OF GENDER ROLES: WOMEN AND THE PLOUGH*

ALBERTO ALESINA
PAOLA GIULIANO
NATHAN NUNN

The study examines the historical origins of existing cross-cultural differences in beliefs and values regarding the appropriate role of women in society. We test the hypothesis that traditional agricultural practices influenced the historical gender division of labor and the evolution of gender norms. We find that, consistent with existing hypotheses, the descendants of societies that traditionally practiced plough agriculture today have less equal gender norms, measured using reported gender-role attitudes and female participation in the workplace, politics, and entrepreneurial activities. Our results hold looking across countries, across districts within countries, and across ethnicities within districts. To test for the importance of cultural persistence, we examine the children of immigrants living in Europe and the United States. We find that even among these individuals, all born and raised in the same country, those with a heritage of traditional plough use exhibit less equal beliefs about gender roles today. *JEL* Codes: D03, J16, N30.

A belief that varies enormously

- World Values Survey, fourth wave, 62 countries surveyed between 1999 and 2004.
- Share agreeing that “when jobs are scarce, men should have more right to a job than women”: **3.6% in Iceland, 99.6% in Egypt.**
- Female labor force participation in 2000 ranged from 16.1% in Pakistan to 90.5% in Burundi.
- Where do differences of that size come from?

Ester Boserup

 16 languages 

Article [Talk](#)

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From Wikipedia, the free encyclopedia

Ester Boserup (18 May 1910^[1] – 24 September 1999) was a Danish [economist](#). She studied economic and agricultural development, worked at the [United Nations](#) as well as other [international organizations](#), and wrote seminal books on [agrarian change](#) and the role of [women in development](#).

Boserup is known for her theory of agricultural intensification, also known as **Boserup's theory**, which posits that population change drives the intensity of agricultural production. Her position countered the [Malthusian theory](#) that agricultural methods determine population via limits on [food supply](#). Her best-known book on this subject, *The Conditions of Agricultural Growth*, presents a "dynamic analysis embracing all types of primitive agriculture." (Boserup, E. 1965. p 13)^[2] A major point of her book is that "[necessity is the mother of invention](#)".

Her other major work, *Woman's Role in Economic Development*, explored the allocation of tasks between men and women, and inaugurated decades of subsequent work connecting issues of gender to those of economic development, pointing out that many economic burdens fell disproportionately on women.^[3] In an early review, her book was called "pioneering;" nearly five decades later, it has proved influential, having been cited by thousands of other works.^[4]

Ester Boserup



Born Ester Børgesen
18 May 1910
[Copenhagen](#), Denmark
Died 24 September 1999 (aged 89)

Boserup's hypothesis

- Ester Boserup (1970): differences in gender roles originate in the form of agriculture traditionally practiced.
- **Shifting cultivation** is labor intensive and uses handheld tools, the hoe and the digging stick.
- **Plough cultivation** is capital intensive, and takes upper body strength, grip strength and bursts of power to pull the plough or control the animal that pulls it.
- Soil preparation is about a third of the total time spent on agricultural tasks. So where the plough is used, men have an advantage in farming.

From a division of labor to a norm

- Men in the fields, women in the home: a specialization of production along gender lines.
- Reinforced by the fact that the plough reduces the need for weeding, a task typically done by women and children.
- Child care fits activities that can be stopped and resumed and that do not put children in danger. Hoeing qualifies. An animal-drawn plough does not.
- The division of labor generated a belief about where women naturally belong, and the belief persists after the economy moves out of agriculture.







Hoe-farming

 7 languages 

Article [Talk](#)

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From Wikipedia, the free encyclopedia

Hoe-farming is a primitive form of [agriculture](#) defined by the absence of the [plough](#). [Tillage](#) in hoe-farming cultures is done by simple manual tools such as [digging sticks](#) or [hoes](#). Hoe-farming is the earliest form of agriculture practiced in the [Neolithic Revolution](#). Early forms of the plough (*[ard](#)*) were introduced throughout the Near East ([Naqada II](#)) and Europe ([Linear Pottery culture](#)) by the 5th to 4th millennium BC. The invention spread throughout [Greater Persia](#) and parts of [Central Asia](#), reaching [East Asia](#) in the 2nd millennium BC ([Chinese Bronze Age](#)).^[1]

The term *hoe-farming* was introduced (as [German](#): *Hackbau*, as opposed to *Ackerbau*) by [Eduard Hahn](#) in 1910.



Use of the [digging stick](#) for tillage in the [Nuba Mountains](#), Southern Sudan (2001 photograph) 



Why a belief would persist

- Cultural beliefs as **rules of thumb**: cheaper than acquiring the information to behave optimally in every instance, and therefore sticky by construction.
- Reinforcement by laws and institutions: property rights, voting rights, parental leave policies.
- Complementarity with industrial structure: beliefs about gender push a society toward capital- and brawn-intensive industries, which lowers the cost of holding the belief.
- The paper's target is persistence that survives once those complementarities are held fixed.

The obvious objection

- Europe and its offshoots have long practiced plough agriculture and look gender-equal today.
- The authors' reply: that presumption is Western-centric. Europe is above average, not at the top.
- Share of national political positions held by women in the United States in 2000: 13%, which ranked it 50th out of 156 countries.
- US female labor force participation the same year: 59.5, 47th out of 181. Both look less equal still once income is taken into account.

The historical data

- The **Ethnographic Atlas**: 1,265 ethnic groups, each coded for the earliest period with satisfactory evidence, and always before industrialization.
- Plough absent for 997 groups, aboriginal for 141, adopted only after European contact for 18.
- There is no evidence of groups switching form of agriculture and switching back. Once adopted, plough use is stable.
- Female participation in agriculture is coded on a five-point scale, from males only to females only.

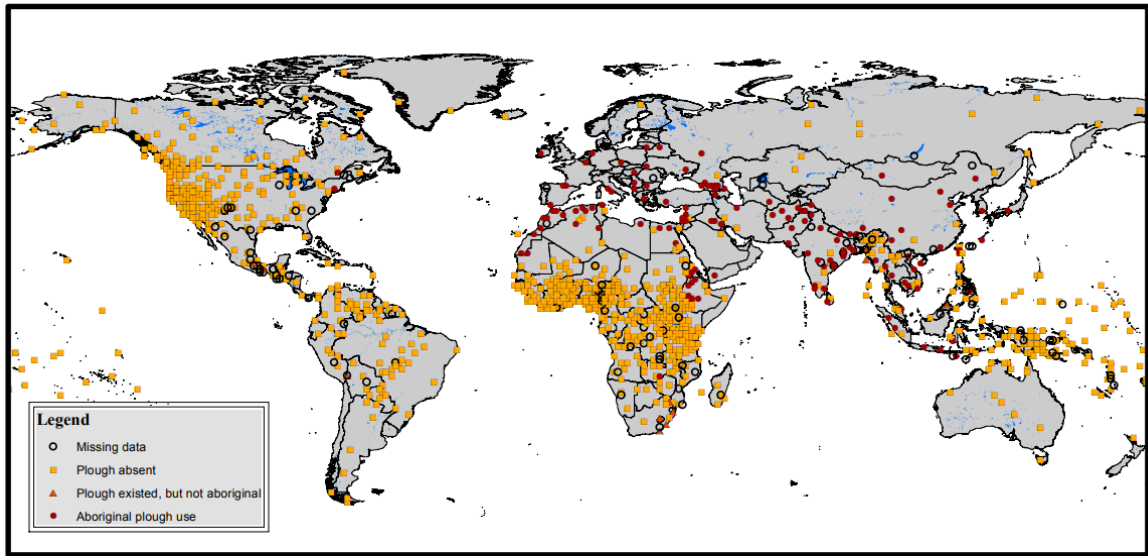


Figure A1: Centroid locations and traditional plough use of ethnic groups within the Ethnographic Atlas.

Plough use across the Atlas

- Each point is an ethnic group at its historical centroid.
- Aboriginal plough use concentrated across Eurasia, the Middle East, North Africa and the Ethiopian highlands.
- Plough absent across most of sub-Saharan Africa, the Americas, island Southeast Asia and Oceania.
- This is the variation the rest of the paper works with.

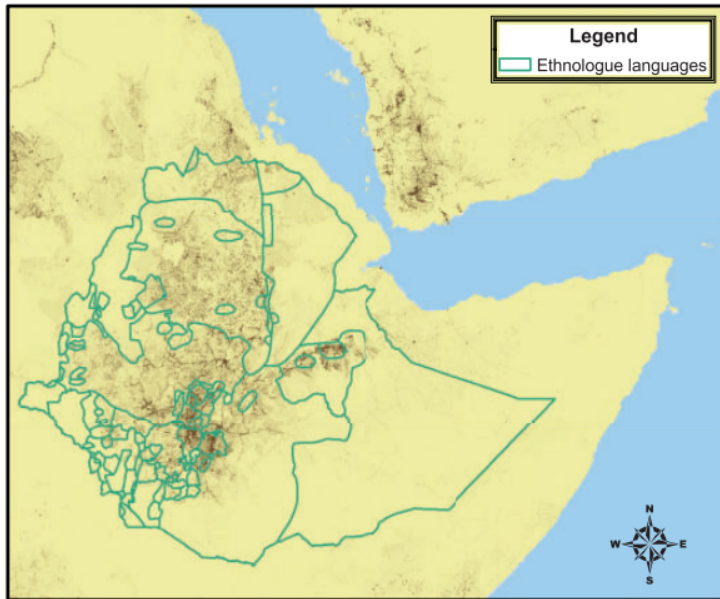
Plough use and women's work, before industry

- Plough use is associated with a fall of **0.88** in female participation in agriculture, on a scale with mean 3.0 and standard deviation 1.0.
- In the Standard Cross-Cultural Sample, 186 societies chosen to be independent of one another, the fall is 1.14, roughly one standard deviation.
- Task by task: falls everywhere, largest in soil preparation, planting and crop tending.
- Outside agriculture, no systematic pattern, except less female participation in burden carrying.

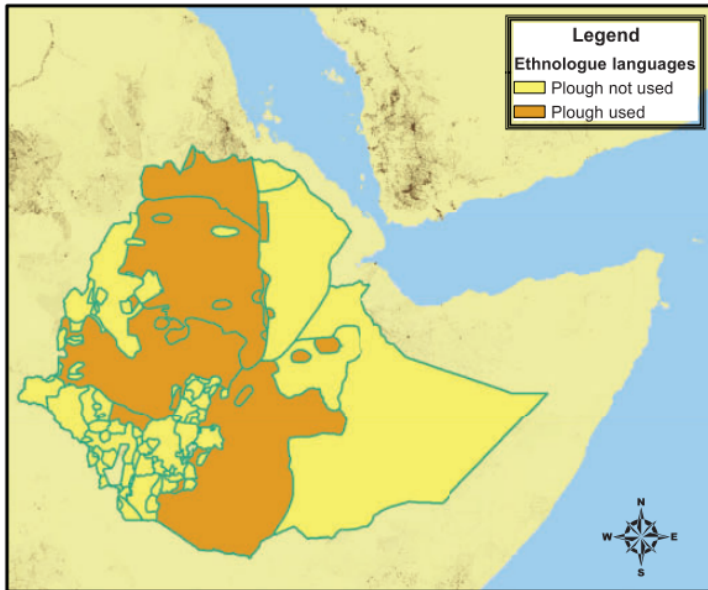
Linking past to present

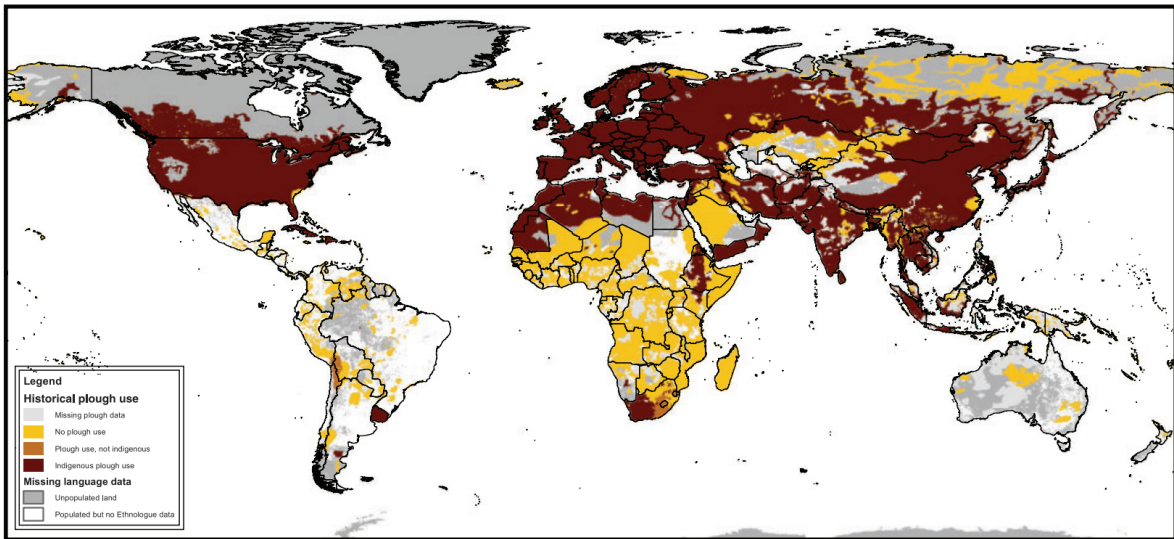
- Plough use is measured by ethnic group. The outcomes are measured by country, by district and by individual.
- Ethnologue's map of 7,612 language groups, matched to the 1,265 Atlas groups, combined with Landsat population at 1 km by 1 km.
- That yields, for any district or country, the **share of today's population whose ancestors used the plough**.
- Ethiopia is the worked example: language group boundaries, population density, and ancestral plough use.

(a) Population density and language groups



(b) Population density, language groups, and their traditional plough use

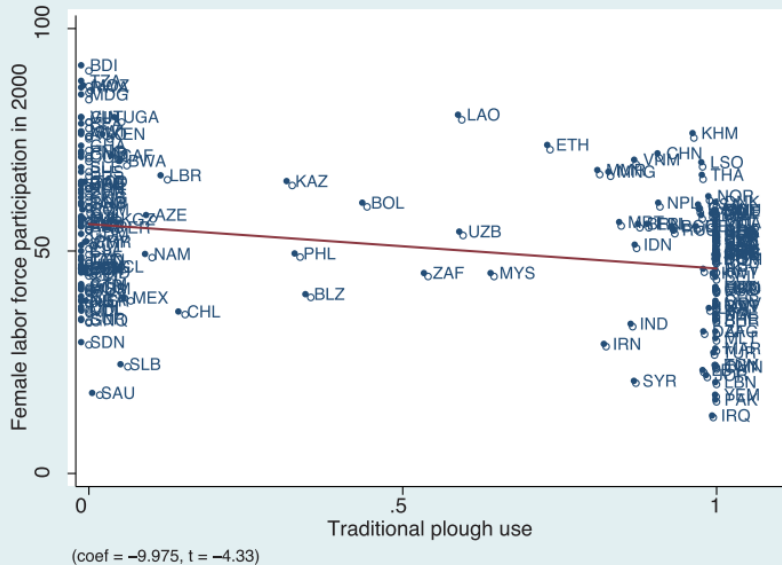




The global picture

- Plough-using ancestry across Eurasia and North Africa, absent across most of sub-Saharan Africa and the Americas.
- Two caveats the authors raise. Much of the variation is at the macro level. And Ethnologue boundaries are missing for parts of Latin America.
- They answer the first by moving inside countries and inside districts, which is where the analysis goes next.

(a) Female labor force participation in 2000



Share of firms with female ownership, 2003–2010

Traditional plough use

(coef = -10.845, t = -3.75)

(coef = -10.845, t = -3.75)

Scatter plot showing the relationship between Traditional plough use (X-axis) and Share of political positions held by women in 2000 (Y-axis). The X-axis ranges from 0 to 1, and the Y-axis ranges from 0 to 40. A positive linear regression line is shown with the equation (coef = 2.291, t = 1.52). Data points are labeled with country codes. Notable outliers include SWE, DEU, BIH, CUB, AUT, TKM, and ISL.

(coef = 2.291, t = 1.52)

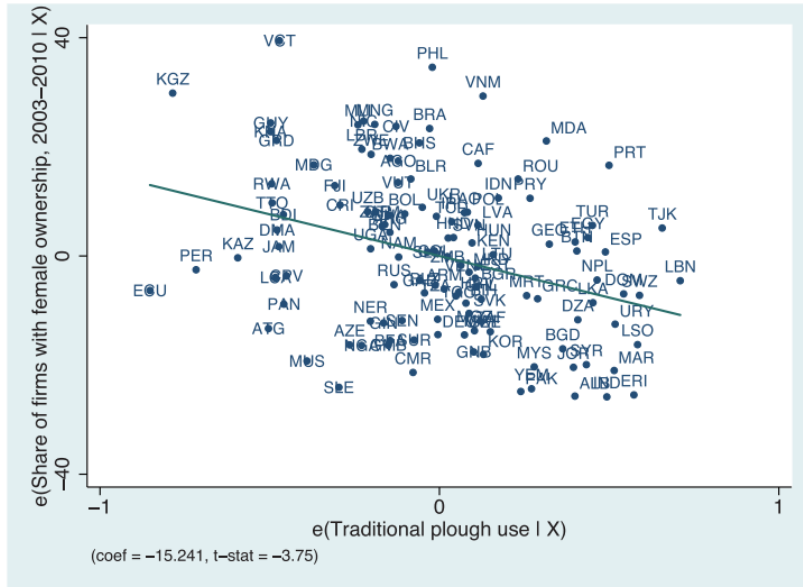
The raw correlations

- Female labor force participation and the share of firms with female ownership both fall with ancestral plough use.
- Female participation in politics: the raw relation is flat and slightly positive, in the wrong direction.
- Once income per capita is taken into account, the political relation roughly doubles and turns negative. Richer countries have more women in politics, and plough countries are richer.

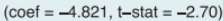
Scatter plot showing the relationship between $e(\text{Female labor force participation in 2000} | X)$ (Y-axis) and $e(\text{Traditional plough use} | X)$ (X-axis). The Y-axis ranges from -40 to 40, and the X-axis ranges from -1 to 1. A negative linear regression line is shown, with a coefficient of -12.401 and a t-stat of -4.18 . The plot includes numerous data points labeled with country codes, indicating a general negative correlation between the two variables.

(coef = -12.401, t-stat = -4.18)

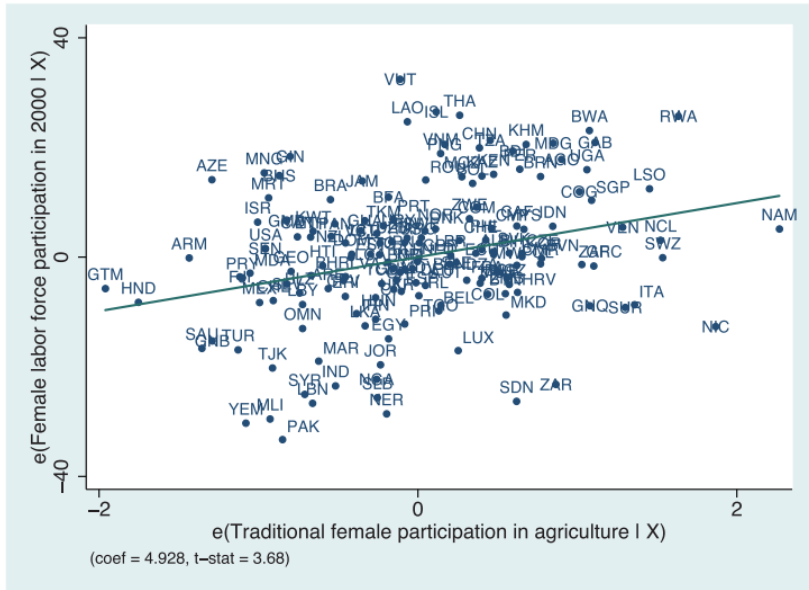
(b) Traditional plough use and current female firm ownership



(c)



(d) Traditional female participation in agriculture and current FLFP



Across countries

- Held fixed: ancestral suitability for agriculture, tropical share, domesticated animals, settlement density, political complexity, and current income.
- A one standard deviation increase in ancestral plough use means **5.85 percentage points** less female labor force participation, 11.4% of the mean.
- 7.19 points fewer firms with female ownership, 20% of that mean. 2.28 points fewer political positions held by women, 19% of that mean.
- Not a sub-Saharan Africa story: the estimates barely move when those countries are dropped.

The last panel

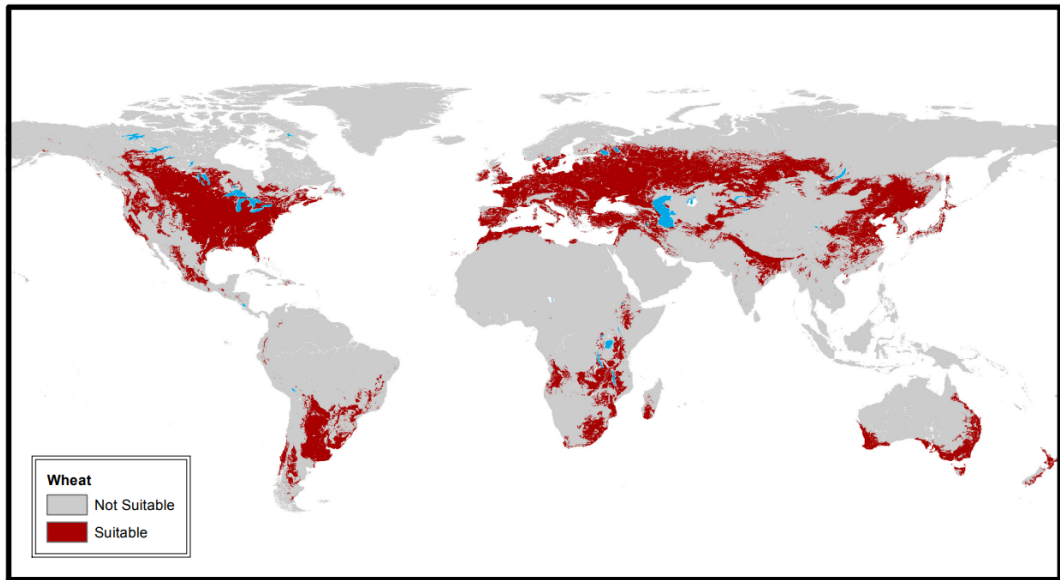
- Female labor force participation today, plotted against women's participation in agriculture in the pre-industrial period.
- Strong persistence across the whole span, from the ethnographic record to the present.

Inside countries

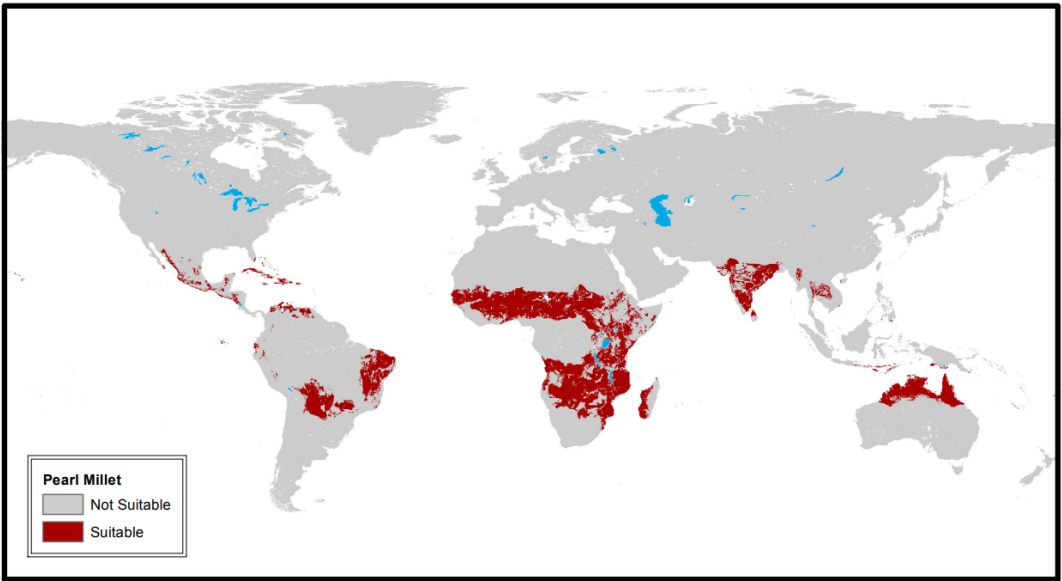
- Individual-level data, comparing people within the same country and then within the same **district**.
- Where ancestors used the plough: less female labor force participation, and more agreement that men have more right to a job and that men make better political leaders.
- Eight censuses with ethnicity codes, from Bolivia and Chile to Nepal and Uganda: the effect on participation runs from 4 to 10 percentage points, averaging 4.
- Roughly half the size of the cross-country magnitudes.

Which came first?

- The worry: societies that already held unequal beliefs may have been the ones to adopt the plough.
- Pryor (1985): the plough pays off for crops needing large tracts prepared in a short period, on soils that are not shallow, sloped or rocky.
- **Plough-positive** cereals: wheat, barley, rye. **Plough-negative**: sorghum, foxtail millet, pearl millet.
- The two sets are otherwise alike. Both cultivated in the Eastern Hemisphere since the Neolithic, both used for flour, porridge, bread or beverages, similar yields, similar population densities.



(a) Wheat suitability



(b) Pearl millet suitability

Suitability, not choice

- Geo-climatic suitability for wheat against pearl millet, from the FAO's global database.
- Where the land suited plough-positive cereals, the plough was adopted, and gender roles are less equal today.
- The estimates hold with overall agricultural suitability, temperature, precipitation, soil depth and slope all held fixed.
- They come out larger than the plain correlations, consistent with historically advanced societies both adopting the plough and being richer today.

Culture, or institutions?

- The long-run effect could run through beliefs, or through institutions, policies and markets that plough agriculture helped build.
- Beliefs are internal to the person. When people move, their beliefs move with them; the external environment stays behind.
- So: **children of immigrants** in the United States and in Europe, all born and raised in the destination country, facing the same laws and markets.
- Their ancestral plough use is taken from the father's country of origin.

The children of immigrants

- Daughters of immigrants to the United States: those with plough-using ancestry participate less in the labor force.
- Children of immigrants in Europe, across three waves of the European Social Survey: less equal beliefs about gender roles.
- Across countries, a one-unit increase in ancestral plough use lowers female labor force participation by 12.4 points. Among daughters of immigrants, only 4.3 to 6.2.
- The authors read that gap as internal norms carrying roughly **35 to 50 percent** of the total effect.

Conclusions

- Descendants of plough societies hold less equal gender norms and show less female participation outside the home, across countries, districts, ethnicities and individuals.
- A sizable part of that runs through **cultural transmission**, and not only through institutions and markets.
- The claim is narrow: it is about whether it is thought as appropriate for women to work outside the home as for men.
- Historical origins do not displace short-run determinants: development, medical improvements, technological change, the structure of production.

In this class

- **Figueroa and Fouka** (2023): industrialization shifted which groups could act on their values, and slavery became repugnant.
- **Dell and Querubín** (2018): firepower applied near civilians raised insurgent activity and weakened the state it was meant to build.
- **Alesina, Giuliano and Nunn** (2013): a farming technology fixed a norm about women's work that outlived the technology by centuries.
- Common thread: material conditions, the distribution of power, and the values that follow from them.